Name: Team Timbuktu **Date:** 3/1/25

Element J: Documentation of External Evaluation

We focused on creating a toy enrichment hanger for an Eland, a sizable species native to Africa. The design was not only ambitious, featuring a 16-foot-tall structure with a 10-foot arm reaching over an 8-foot fence, but it also required careful consideration of safety and operability. To ensure that our design would effectively meet the needs of the animal while adhering to safety guidelines, we consulted with a variety of stakeholders and experts throughout the process. Their insights were invaluable, allowing us to adapt and enhance our project effectively. As stakeholders in our project, we included members from the local zoo, animal welfare organizations, and our academic advisors. Each group offered a unique perspective that contributed to our understanding of the Eland's behavior and environmental requirements. For instance, the local zoo staff provided us with practical insights into how similar structures function in their facilities, emphasizing the importance of stability and safety in any enrichment device.

By collaborating with these stakeholders, we gathered comprehensive information and guidance that proved essential throughout our development process. Once we had constructed our initial design, we conducted a thorough evaluation to assess its functionality and safety. This evaluation process involved reviewing the structural integrity of the hanger, particularly focusing on the arm attachment, which was a crucial aspect of the design. We conducted stress tests to simulate the forces exerted during use, ensuring that the arm could withstand the weight of toys and other enrichment items without compromising safety. This comprehensive analysis revealed potential weaknesses that needed to be addressed before final implementation. Towards the end of our construction, we reached out to field experts for their evaluation of our prototype. Their feedback highlighted the need for enhanced support at the joint

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where the arm attaches to the main structure. Specifically, they recommended integrating an L bracket on the backside of the arm to provide additional stability and to prevent unwanted rotation. This insight validated our own concerns regarding the joint's stability and prompted us to refine our design further, ensuring that it would endure the rigors of use with an active Eland.

In synthesizing the project's external evaluation and the feedback we received, we recognized the critical importance of incorporating expert advice and stakeholder input into our design process. By actively engaging with those who possess extensive knowledge of animal behavior and enrichment solutions, we were able to identify potential pitfalls and enhance the overall quality of our project. The recommendations made by experts on the needed modifications not only supported our aims but also reinforced our commitment to crafting a safe and enriching environment for the Eland. The outcome of this project ultimately reflects a collaborative effort that brought together diverse expertise aimed at enriching the life of the Eland. Our toy enrichment hanger, now reinforced with additional support as per expert recommendations, represents both our dedication to animal welfare and our growth as a team. This experience has taught us that the involvement of stakeholders and field experts is paramount in the innovation process, revealing that successful projects are built on a foundation of collaboration, ongoing evaluation, and a willingness to adapt based on expert feedback. As we look forward to the installation of our hanger, we remain committed to continuous improvement, ensuring that our design not only meets but exceeds the requirements for enriching the Eland's habitat.